Test Specification (검증용 샘플 문서) 1

Test Sample 1. Temperature Shock Resistance Test

Purpose

When the DUT is exposed to rapid temperature changes, it shall maintain its operational integrity without functional degradation. The component must demonstrate stable performance across extreme temperature transitions.

After exposure to temperature shock cycles, the DUT shall automatically resume normal operation within the specified temperature range without requiring manual intervention.

Any temporary performance limitations during transition phases shall be documented in the technical specifications.

Test

Parameter	Specification
Operating mode	Operating mode III.b
Test duration	Temperature hold time: t_hold = 30 min per step
Test temperature	Low temperature: T_low = -40°C High temperature: T_high = +85°C

Parameter	Specification
	Transition time: ≤ 3 minutes
Test voltage	13.5V
No. of cycles	50 cycles

Temperature cycling shall proceed as follows:

- Start at ambient temperature (25°C ± 5°C)
- Ramp to T_high and hold for 30 minutes
- Rapid transition to T_low within 3 minutes
- Hold at T_low for 30 minutes
- Return to T_high

A maximum temperature deviation of ±3°C shall be maintained during hold periods.

Requirement

After the test, the DUT shall automatically return to operating mode III.b with no functional errors. Visual inspection shall confirm no physical damage.

분류: Environmental Test

간단 설명: 급격한 온도 변화에 대한 센서 모듈의 내구성 평가

동작 기준:

- 저온(-40°C)과 고온(+85°C) 구간을 반복적으로 전환하며 동작 상태 확인
- 각 온도 구간에서 30분간 유지하며 기능 이상 여부 모니터링

Test Sample 2. Electromagnetic Compatibility (EMC) Test

Purpose

The DUT shall demonstrate immunity to electromagnetic interference and shall not emit electromagnetic radiation exceeding specified limits. This ensures the component operates reliably in the vehicle's electromagnetic environment without causing interference to other systems.

Test procedure

Emission Test:

- Conducted emissions: 150 kHz to 108 MHz
- Radiated emissions: 30 MHz to 1 GHz
- Test according to CISPR 25 Class 3

Immunity Test:

- Bulk Current Injection (BCI): 1 MHz to 400 MHz
- Radiated immunity: 80 MHz to 6 GHz, field strength 30 V/m
- ESD: ±8 kV contact discharge, ±15 kV air discharge

Test duration:

Approx. 48 hours total

- Emission testing: 16 hours
- Immunity testing: 32 hours

All tests shall be performed at operating mode III.a with V_test = 13.5V and at ambient temperature.

Requirement

Emission levels shall not exceed limits specified in CISPR 25 Class 3. During immunity testing, the DUT shall maintain full functionality with no performance degradation or data corruption.

분류: EMC Test / Electrical Test

간단 설명: 전자기 적합성 평가를 통한 방사 및 전도 내성 확인

시험 조건:

- 방출 시험: CISPR 25 Class 3 기준 적용
- 내성 시험: BCI, 방사 내성, ESD 시험 포함
- 동작 전압 13.5V에서 전 시험 수행

Test Sample 3. Salt Spray Corrosion Test

Test

The test shall be performed in accordance with ISO 9227 with the following parameters:

Parameter	Specification
Operating mode	Non-operating (storage condition)
Test duration	240 hours continuous exposure
Salt solution	5% NaCl solution, pH 6.5-7.2
Chamber temperature	35°C ± 2°C
No. of cycles	1 cycle (continuous)

Test Procedure:

1. Pre-conditioning: Clean and dry DUT, measure initial resistance

2. Exposure: Place DUT in salt spray chamber for 240 hours
3. Post-conditioning: Rinse with deionized water, dry at room temperature for 24 hours
4. Evaluation: Visual inspection and electrical performance verification

Requirement

The corrosion protection level specified in the component specifications shall be achieved. No red rust formation on critical surfaces. Electrical performance degradation shall not exceed 5% of initial values.

분류: 부식성 평가

간단 설명:

ISO 9227 규격 기반 염수 분무 시험으로 부식 저항성 평가.

Daeyang Sensor Module은 일반적으로 240시간 연속 노출 조건 적용.

시험 후 전기적 성능 저하가 초기 대비 5% 이내여야 하며, 주요 표면에 적청 발생이 없어야 함.

Test Sample 4. Mechanical Shock Test

Test

Test shall be conducted according to the Mechanical Shock test described in ISO 16750-3, section 4.2.1.

Parameter	Specification
Operating mode	Operating mode III.b
Test duration	3 shocks per direction, 18 shocks total
Test voltage	12V
Shock profile	Half-sine pulse
Peak acceleration	50 g
Pulse duration	11 ms

Test Procedure:

Shocks shall be applied in all three orthogonal axes (X, Y, Z), both positive and negative directions. The DUT shall be operating during shock application.

Requirement

There shall be no mechanical damage, loose parts, or functional interruption during or after the test. The performance specifications must be satisfied. Electrical continuity shall be maintained throughout the test.

분류: 기계적 충격 내구성 평가

간단 설명:

ISO 16750-3 규격 기반 기계적 충격 시험으로 충격 저항성 평가.

Half-sine 파형, 50g 가속도, 11ms 지속 시간 조건으로 3축 6방향 각 3회 충격 인가.

시험 중 전기적 연속성 유지 및 기능 중단 없어야 함.

추가 정보

충격 시험 파형 종류:

- Half-sine pulse / Trapezoidal pulse / Sawtooth pulse